

Paper D-10: Information Flow: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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Abstract

This paper treats digital information flow as a CDFD/AFL modelling hypothesis. Posts, shares, links, semantic updates, and attention are represented as candidate fluxes Φ moving through cognitive, algorithmic, and network constraints C . The goal is not to certify a social-media model. The goal is to state measurable variables and failure conditions for overload, echo chambers, viral cascades, and semantic bottlenecks. Release-local runtime outputs are internal diagnostics and tests that could break the mapping, not field evidence.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, and Λ denotes the Life Number where used. The coefficients α , β , and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or redistribution.

1 Information as Constrained Transport

Digital networks move information quickly, but not freely. Human attention, platform ranking, moderation queues, recommendation systems, language boundaries, and trust relationships all act as constraints. CDFD/AFL frames the system as a transport problem: when information flux rises faster than attention and routing constraints can relax, the system can enter overload, fragmentation, or repetitive circulation.

2 Candidate Variable Map

3 Echo Chambers and Viral Cascades

An echo chamber is treated here as a candidate low-constraint loop inside a larger high-constraint network. A viral cascade is treated as a local flux surge that exceeds available attention and routing capacity. These are modelling hypotheses. They need platform

CDFD term	Information-flow reading	Example proxy
Φ	semantic or attention flux	posts, shares, impressions, messages
C	attention or routing constraint	ranking friction, moderation delay, trust barrier
S	responsiveness	fact checking, topic diversity, cross-cluster routing
M_s	memory	prior beliefs, platform history, cached ranking state
Ψ_s	operating ratio	overload, virality, isolation, or balanced flow

data, time scales, and comparison against standard network-science baselines before they become empirical claims.

4 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system’s ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

5 Current Runtime Diagnostic

The release-local domain adapter sweep includes an information-theory row and related network-routing rows. Those artifacts test whether the current runtime returns finite outputs and interpretable regimes under toy conditions. They do not validate polarization, virality, or platform-behavior claims.

6 Falsification Targets

The information-flow mapping weakens if measured attention and routing constraints do not track overload, if echo-chamber formation is better explained without CDFD/AFL variables, or if the model cannot distinguish between semantic content, network topology, and platform policy effects.

7 Conclusion

Information flow can be studied as constrained transport when the variables are defined carefully. In Part IV, the CDFD/AFL contribution is an audit language for candidate overload and memory effects, not a proof of social behavior.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs` and `Part_E_Synthesis/figures/`.

8 Reference Layer

References